USC University of Southern California

Quantum Monte Carlo without the guesswork

A permutation matrix representation approach

Nic Ezzell, Lev Barash, and Itay Hen

USC Physics & Astronomy
USC Information Sciences Institute

About me



Mississippi School for Math & Science



Mississippi State University



USC DOE CSGF



HRL Laboratories



Quantum Elements

My work: Quantum computation and information ▪ Scientific computing

More: blogs, papers, and related projects `naezzell.github.io`

Motivation

Complicated models and their phase transitions

Goal: specify a Hamiltonian, measure its response, and locate phase transitions.

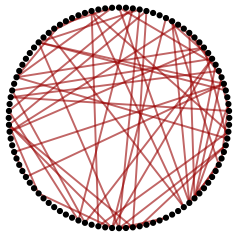
A controlled test on 100 spins

$$H_U(\lambda) = U\big[(H_0 + \lambda H_1) \otimes I_{98}\big]U^\dagger,$$

$$H_0 = Z_1Z_2 + 0.1(X_1 + X_2), \qquad H_1 = Z_1 + Z_2.$$

I_{98} is the identity on 98 additional spins.

$U = \sum_{a=1}^{L} c_a Q_a$, with real c_a, $\sum_a c_a^2 = 1$, and mutually anticommuting Pauli strings Q_a.



100 spins, nonlocal couplings.
Schematic connections.

Advanced-measurement example: 59 terms in U give 179 Pauli terms in $H_U(1)$. Its rotated observable has 295 terms.

The rotation makes the input complicated while keeping an exact benchmark.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4, Appendix D. Ezzell & Hen, arXiv:2504.07295v2. Sec. IX C.

Using our code is very easy

Define H as a text file

```
-1.00 1 Z 2 Z 3 Z
-0.68 1 X
-0.73 1 Z 2 X
 0.82 1 Y 3 Y
```

Define sim params

```
#define beta 2.0
#define steps 100000
```

Choose what to estimate

```
#define MEASURE_HDIAG
#define MEASURE_HDIAG_CORR
#define MEASURE_HDIAG_EINT
#define MEASURE_HDIAG_FINT
```

Run sim and analyze

```
Number of MPI processes: 6

Collecting statistics and finalizing the calculation

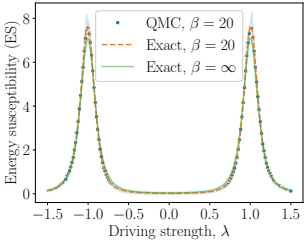
Total number of MC updates = 600000
Total mean(q) = 3.365
Total max(q) = 14
Total mean(sgn(W)) = 0.674666667
Total std.dev.(sgn(W)) = 0.00939374389
Total of observable #1: H_{diag}
Total mean(O) = -0.800449775
Total std.dev.(O) = 0.0133078895
```

Open source code and data:
https://zenodo.org/records/15787829
GitHub: source code (v2.1.0)

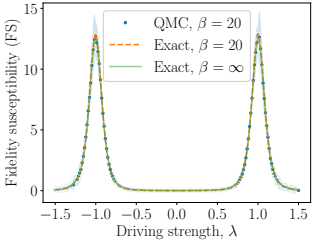
```
Observable #1: H_{diag}, mean of std.dev.(O) =
0.0353, std.dev. of mean(O) = 0.0409: test passed
```

PMR-QMC reproduces the random ensemble's response

Energy susceptibility



Fidelity susceptibility



100 spins, $\beta = 20$: QMC matches the exact finite-temperature curves and resolves both response peaks near $\lambda = \pm 1$.

Averages over 10 models selected for favourable average signs.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4(b,c). IUPAP CCP 2025, printed slide 33.

QMC implementations often depend on the model

A widely used method, **stochastic series expansion (SSE)**, illustrates the problem:

Spin-$\frac{1}{2}$ model	**Model-specific SSE method**
Heisenberg magnet	A. W. Sandvik, Phys. Rev. B **56**, 11678 (1997).
Transverse-field Ising	A. W. Sandvik, Phys. Rev. E **68**, 056701 (2003).
Rydberg atom arrays	E. Merali, I. J. S. De Vlugt & R. G. Melko, SciPost Phys. Core **7**, 016 (2024).

Changing the interactions can require new choices of how to split the Hamiltonian into elementary terms and how to sample them.

Our goal: change model and observable inputs, while reusing the code.

Melko, *Stochastic Series Expansion QMC* (2013). Ezzell & Hen, arXiv:2504.07295v2. Introduction.

The ingredients behind PMR-QMC

1. **A group decomposition of matrices**
   Write $H = \sum_j D_j P_j$: diagonal weights times permutations from a group acting regularly on basis states.
2. **An expansion of the partition function**
   Sum closed permutation paths. Divided differences encode the diagonal energies along each path.
3. **Automated QMC code generation**
   Use (1) and (2) to construct the Hamiltonian representation and sampling updates from the model input.
4. **Estimators for arbitrary observables**
   Combine (1) and (2) with new divided-difference identities to derive static, imaginary-time, and integrated-response estimators.

A common mathematical framework for model input, sampling, and measurement.

Barash, Babakhani & Hen, arXiv:2307.06503v2. Ezzell, Barash & Hen, arXiv:2408.03924v3. Ezzell & Hen, arXiv:2504.07295v2.

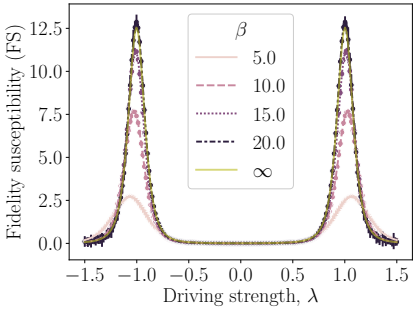
Preview results

Simple two qubit PRL model

A simple two-spin model:

$$H(\lambda) \equiv (H_0 + \lambda H_1),$$
$$H_0 \equiv Z_1 Z_2 + 0.1(X_1 + X_2),$$
$$H_1 \equiv Z_1 + Z_2$$

Fidelity susceptibility



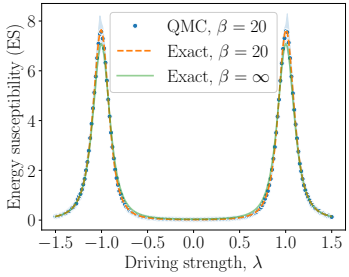
[†] Ezzell, Barash & Hen, arXiv:2408.03924v3.

A random ensemble model

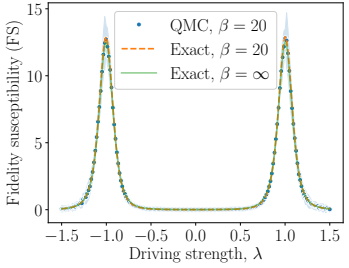
$$H(\lambda) \equiv (H_0 + \lambda H_1),$$
$$H_0 \equiv Z_1 Z_2 + 0.1(X_1 + X_2),$$
$$H_1 \equiv Z_1 + Z_2$$

$$H_{U_i}(\lambda) \equiv U_i(H_0 + \lambda H_1)U_i^\dagger$$

Energy susceptibility



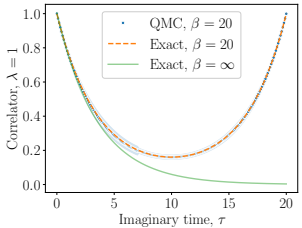
Fidelity susceptibility



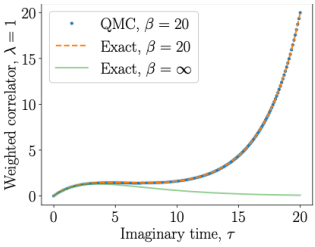
[†] Ezzell, Barash & Hen, arXiv:2408.03924v3.

PMR-QMC also measures imaginary-time correlations

Connected correlator $G(\tau)$



Weighted correlator $\tau G(\tau)$



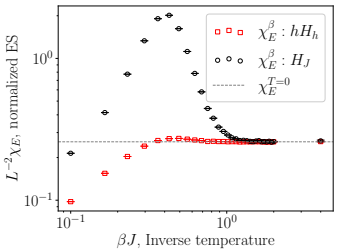
$G(\tau) = \langle A(\tau)A\rangle - \langle A\rangle^2$, with $A = \partial_\lambda H_U$.

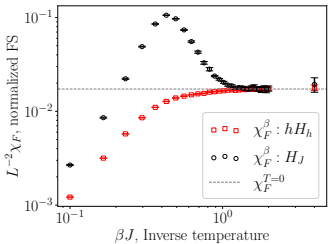
The same simulation measures the curves whose integrals give the susceptibilities.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4(a). Weighted panel: IUPAP CCP 2025, printed slide 34.

TFIM on a square lattice results

$$H_{\mathrm{TFIM}}(h) = H_J + hH_h = -\sum_{\langle i,j\rangle} X_i X_j - h\sum_{i=1}^{n} Z_i.$$

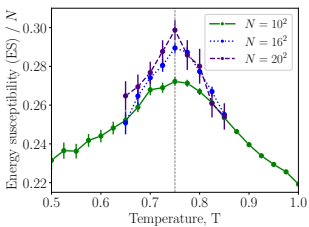


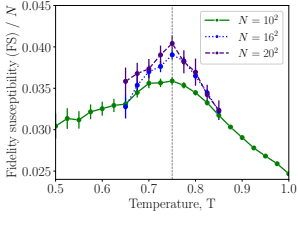


[*] Ezzell, Barash & Hen, arXiv:2408.03924v3.
[**] Albuquerque *et al.*, Phys. Rev. B **81**, 064418 (2010).

XXZ on a square lattice results

$$H = \frac{J_z}{4} \sum_{\langle i,j \rangle} Z_i Z_j + \frac{\lambda}{4} \sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j)$$



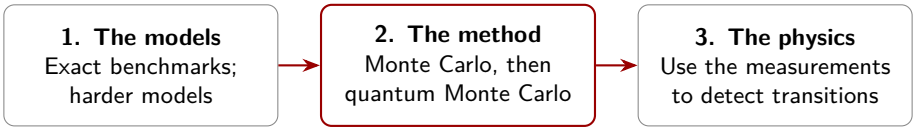


[*] Ezzell, Barash & Hen, arXiv:2408.03924v3.
[**] L. Wang, Y.-H. Liu, J. Imriška, P. N. Ma, and M. Troyer, Fidelity Susceptibility Made Simple: A Unified Quantum Monte Carlo Approach, Phys. Rev. X 5, 031007 (2015).

Background: magnets and Monte Carlo

From solvable magnets to numerical simulation

How do we study a quantum magnet when no analytic solution is available?



Start with a 2D classical magnet and a 1D quantum chain: two solvable benchmarks.

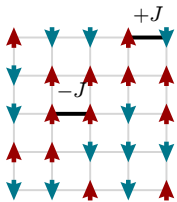
Sampling lets us move beyond the special models we can solve on paper.

Classical phase transitions

Start with a classical magnet

On a **2D square lattice**, each spin is a number $z_i = \pm 1$:

$$E(z) = -J \sum_{\langle i,j \rangle} z_i z_j, \qquad J > 0.$$



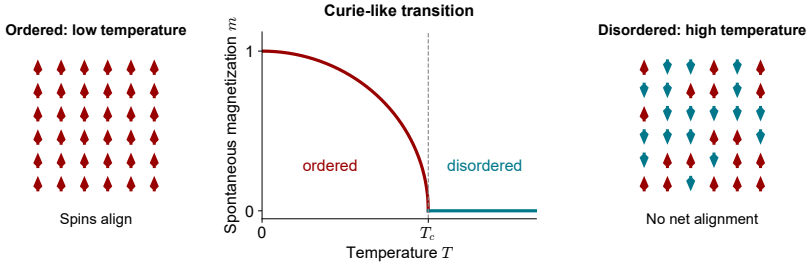
Energy favours alignment. Each aligned bond contributes $-J$; an opposite pair contributes $+J$.

Temperature allows disorder. A pattern has probability

$$p(z) = \frac{e^{-\beta E(z)}}{\mathcal{Z}}, \qquad \mathcal{Z} = \sum_z e^{-\beta E(z)}, \quad \beta = 1/T.$$

The competition between energy and entropy can change the phase.

Heating a magnet destroys its order



Schematic curve and spin cartoons; thermodynamic limit, selected magnetization branch.

The **order parameter** is the spontaneous magnetization: $m = N^{-1} \sum_i \langle z_i \rangle$, nonzero below T_c and zero above it.

A Curie-like transition: thermal disorder overcomes alignment.

A quantum Ising chain

From a 2D classical magnet to a 1D quantum chain

The **1D transverse-field Ising model (TFIM)** adds a quantum spin-flipping field:

$$H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i, \qquad J, h > 0.$$

The interaction favours alignment

$$Z\,|\uparrow\rangle = |\uparrow\rangle\,, \qquad Z\,|\downarrow\rangle = -\,|\downarrow\rangle\,.$$

Low-energy spin patterns look like the classical ferromagnet.

The field favours a superposition

$$X\,|\uparrow\rangle = |\downarrow\rangle\,, \qquad X\,|\downarrow\rangle = |\uparrow\rangle\,.$$

Its preferred single-spin state is $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$.

These competing terms do not commute: tuning h/J changes the ground state.

X, Z are Pauli matrices; $\hbar = k_B = 1$ throughout.

At zero temperature, turning a knob changes the phase

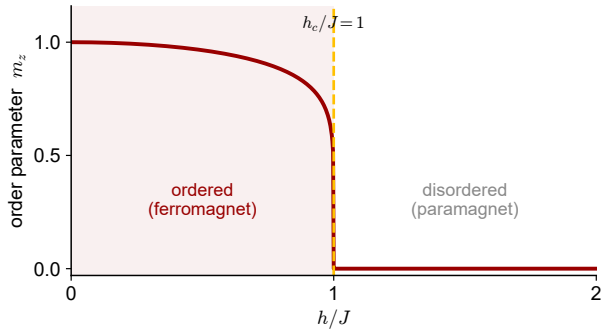
$h/J \to 0$

Ising ferromagnet

$|\uparrow \cdots \uparrow\rangle$

or

$|\downarrow \cdots \downarrow\rangle$



$h/J \to \infty$

quantum paramagnet

$|+\rangle^{\otimes N}$

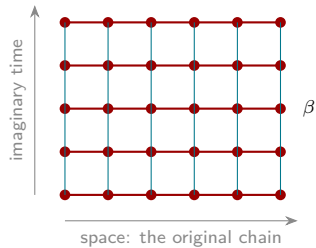
The order parameter is $m_z = \langle Z_i \rangle$ in a selected ordered phase. Exact 1D chain: $h_c/J = 1$ and $m_z = [1 - (h/J)^2]^{1/8}$ for $h < J$. The knob is a coupling; the temperature is zero.

Imaginary time supplies the extra dimension

Split $\mathcal{Z} = \mathrm{Tr}\, e^{-\beta H}$ into many short **imaginary-time steps**.

The Suzuki–Trotter mapping

$$\underbrace{D \text{ spatial dimensions} + 1 \text{ time direction}}_{(D+1)\text{-dimensional classical model}}$$



For the TFIM:

1D quantum chain $\longleftrightarrow$

anisotropic 2D classical Ising model.

The extra direction has extent $\beta = 1/T$; the mapping is exact as the time step goes to zero.

Suzuki's mapping (1976) opened a route to quantum-spin Monte Carlo (1977).

Suzuki, PTP **56**, 1454 (1976); Suzuki, Miyashita & Kuroda, PTP **58**, 1377 (1977).

Beyond solvable models: why we need simulation

The 2D classical and 1D quantum solutions give us benchmarks. Higher dimensions give us important models without a comparable general exact solution.

3D classical Ising magnet

$$E(z) = -J \sum_{\langle i,j \rangle} z_i z_j$$

Neighbours on a cubic lattice.

Use classical Monte Carlo to estimate thermal averages and locate the thermal transition.

2D quantum Ising magnet

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_i X_i$$

Neighbours on a square lattice.

Use quantum Monte Carlo to measure equilibrium properties and locate the field-driven transition.

We want energies, correlations and phase boundaries without an analytic solution.

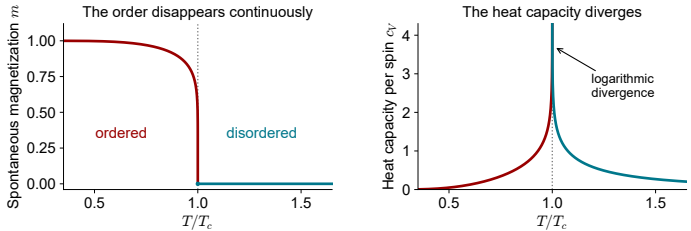
3D Ising: arXiv:1806.03558. 2D TFIM: Albuquerque *et al.*, Phys. Rev. B **81**, 064418 (2010).

Beyond order parameters

The Ising transition also appears in the heat capacity

Return to our **2D classical magnet**: use its exact solution to test the measurements.

$$f = \frac{F}{N} = -\frac{T}{N}\ln \mathcal{Z}, \qquad c_V = \frac{C}{N} = -T\frac{\partial^2 f}{\partial T^2}.$$



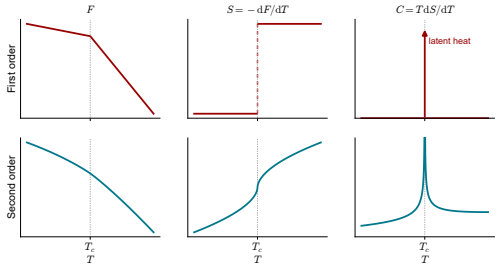
Exact infinite square lattice, zero field: $T_c/J = 2/\ln(1+\sqrt{2}) \simeq 2.269$.

At the same T_c: magnetization and entropy are continuous, but heat capacity diverges.

Onsager, Phys. Rev. **65**, 117 (1944); Yang, Phys. Rev. **85**, 808 (1952).

Beyond Ising: the Ehrenfest classification

Ehrenfest: a transition is nth order if the nth derivative of the free energy F is the first to be discontinuous.



First order: $S = -\partial_T F$ jumps. **Continuous:** S is continuous.

The Ising example belongs to the continuous case: its second derivative diverges.

Schematics. At $T > 0$, singularities require the thermodynamic limit.

At zero temperature, follow the ground-state energy

$$H(\lambda) = H_0 + \lambda H_1, \qquad F = -\frac{1}{\beta} \ln \sum_n e^{-\beta E_n} \xrightarrow{\beta \to \infty} E_0.$$

- The tuning parameter is now a **coupling** λ, not temperature.
- A quantum phase transition changes the ground-state phase as λ is varied in the thermodynamic limit.
- Nonanalytic behaviour of the energy density E_0/N and its derivatives provides one diagnostic.

Finite systems usually show smooth crossovers; follow how they evolve with size.

Exact ground-state level crossings can occur even at finite N.

Sachdev, *Quantum Phase Transitions*, 2nd ed. (2011).

Probe 1: how strongly does the energy bend?

For $H(\lambda) = H_0 + \lambda H_1$, with a smooth, nondegenerate ground state:

$$\frac{\partial E_0}{\partial \lambda} = \langle H_1 \rangle_0 \,, \qquad \boxed{\chi_E = -\frac{\partial^2 E_0}{\partial \lambda^2}}.$$

- **Slope:** the expectation value of the driving term (Hellmann–Feynman).
- **Curvature:** how rapidly that expectation changes.
- A jump in the slope indicates first-order behaviour; singular curvature can signal a continuous transition.

Energy susceptibility measures the response to the chosen drive.

Probe 2: how quickly does the ground state change?

Compare ground states at nearby couplings using their **fidelity**:

$$\mathcal{F}(\lambda, \delta\lambda) = |\,\langle\psi_0(\lambda)|\psi_0(\lambda+\delta\lambda)\rangle\,| = 1 - \tfrac{1}{2}\chi_F(\delta\lambda)^2 + \cdots .$$

- $\mathcal{F} \approx 1$: the two states are nearly the same.
- Large χ_F: a small parameter change strongly changes the state.
- The absolute overlap removes the arbitrary overall quantum phase.

Fidelity susceptibility probes state change without choosing a local order parameter.

A peak is a clue; size and temperature dependence must still be checked.

Zanardi & Paunković, Phys. Rev. E **74**, 031123 (2006).

Small gaps matter when the drive couples to them

Let $\Delta_n = E_n - E_0 > 0$ and $M_n = \langle \psi_n | \, H_1 \, | \psi_0 \rangle$.

Energy susceptibility

$$\chi_E = 2 \sum_{n \neq 0} \frac{|M_n|^2}{\Delta_n}$$

One inverse power of the gap.

Fidelity susceptibility

$$\chi_F = \sum_{n \neq 0} \frac{|M_n|^2}{\Delta_n^2}$$

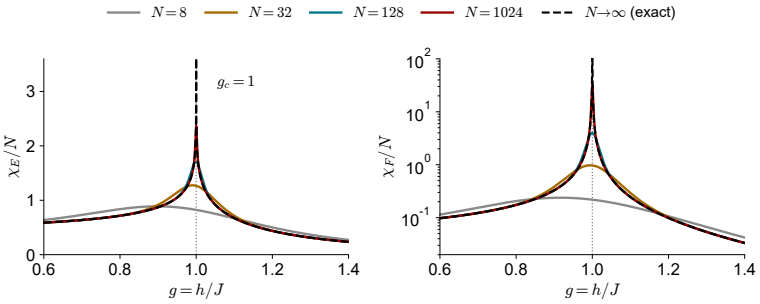
An extra inverse power emphasizes small gaps.

A small gap contributes only when the drive couples to that excitation: $M_n \neq 0$.

Fidelity can reveal a stronger singularity, but neither probe responds to every gap.

Albuquerque *et al.*, Phys. Rev. B **81**, 064418 (2010).

Larger Ising chains approach the exact critical point



Exact periodic chains; $J=1$, $T=0$. Dashed: infinite chain. Fidelity uses a logarithmic scale.

Away from $g=h/J=1$, the curves converge to the infinite-chain response.

The peak approaches the exact $g_c=1$; its height grows without bound.

Damski & Rams, J. Phys. A **47**, 025303 (2014), arXiv:1308.5917.

The bridge to response: imaginary-time correlations

For $H(\lambda) = H_0 + \lambda H_1$, define

$$H_1(\tau) = e^{\tau H} H_1 e^{-\tau H}, \qquad G_0(\tau) = \langle H_1(\tau) H_1 \rangle_0 - \langle H_1 \rangle_0^2 .$$

- τ is **imaginary time**: evolution uses exponentials without i.
- Subtracting the mean product keeps the **connected** response.
- Inserting energy eigenstates gives

$$G_0(\tau) = \sum_{n \neq 0} |M_n|^2 e^{-\tau \Delta_n}, \qquad M_n = \langle \psi_n | \, H_1 \, | \psi_0 \rangle .$$

Small gaps decay slowly: low-energy response lives at long imaginary times.

Integrate the correlation to get both probes

For a positive gap Δ,

$$\int_0^\infty e^{-\tau\Delta}\,\mathrm{d}\tau = \frac{1}{\Delta}, \qquad \int_0^\infty \tau e^{-\tau\Delta}\,\mathrm{d}\tau = \frac{1}{\Delta^2}.$$

The spectral sums become correlation integrals

$$\chi_E = 2\int_0^\infty G_0(\tau)\,\mathrm{d}\tau, \qquad \chi_F = \int_0^\infty \tau G_0(\tau)\,\mathrm{d}\tau.$$

The extra factor of τ emphasizes the slowly decaying, low-energy part.

QMC can measure the response through correlations, without computing eigenstates.

Albuquerque *et al.*, Phys. Rev. B **81**, 064418 (2010).

Lower temperature means more imaginary time

Use the thermal connected correlator $G_\beta(\tau) = \langle H_1(\tau) H_1 \rangle_\beta - \langle H_1 \rangle_\beta^2$:

$$\chi_E(\beta) = \int_0^\beta G_\beta(\tau)\,\mathrm{d}\tau, \qquad \chi_F(\beta) = \int_0^{\beta/2} \tau G_\beta(\tau)\,\mathrm{d}\tau.$$

- $\beta = 1/T$ sets the available imaginary-time interval.
- Increase β until the observables converge at each system size.
- Then compare sizes to distinguish a crossover from critical scaling.

Larger systems near criticality generally require lower temperatures.

This QMC definition of $\chi_F(\beta)$ shares the ground-state limit with fidelity; it generally differs from the finite-temperature Bures metric.

Albuquerque *et al.*, Phys. Rev. B **81**, 064418 (2010). Sec. II B.

From a change of phase to a measurable response

1. **Identify the tuning parameter:** $H(\lambda) = H_0 + \lambda H_1$.
2. **Choose a probe:** order, energy curvature, or fidelity susceptibility.
3. **Measure with QMC:** sample configurations and estimate correlations.
4. **Establish the limit:** check errors, temperature convergence and size scaling.

The bridge is a correlation function: it turns ground-state sensitivity into a sampling problem.

Where PMR enters: constructing broadly applicable weights and response estimators is the next step beyond this background.

Ezzell, Barash & Hen, arXiv:2408.03924; Ezzell & Hen, arXiv:2504.07295.

From Monte Carlo to quantum Monte Carlo

The obstacle: exponentially many states

Without an analytic shortcut, the classical sum contains 2^N spin patterns; the quantum Hilbert space has the same dimension.

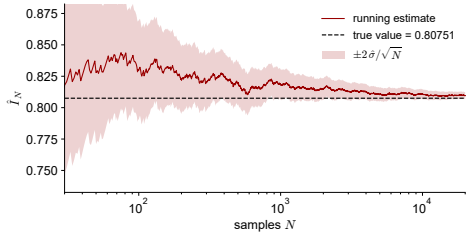
Spins	States
16	$65{,}536$
64	1.8×10^{19}
100	1.3×10^{30}

At a billion terms per second, summing 2^{100} terms takes about 4×10^{13} years.

Monte Carlo estimates an average by visiting a small fraction of the states.

Monte Carlo: replace a sum or integral by samples

$$I = \int_0^1 e^{-x^3}\,\mathrm{d}x = \mathbb{E}_{x\sim\mathcal{U}[0,1]}[e^{-x^3}] \approx \frac{1}{N_s}\sum_{k=1}^{N_s} e^{-x_k^3}.$$



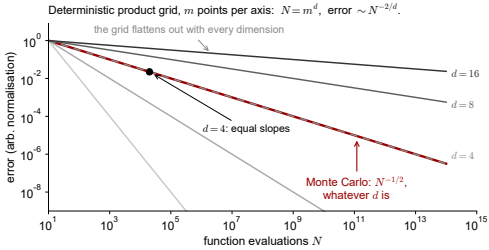
Each sampled point contributes one function value to the average.

Independent samples give error $\sigma/\sqrt{N_s}$: four times the samples halves it.

Monte Carlo also works in high dimensions

Our one-variable example extends to d variables with the same sample mean.

A product trapezoid grid uses $N_s = m^d$ points; for smooth integrands, error $O(N_s^{-2/d})$.



Monte Carlo keeps its $N_s^{-1/2}$ convergence rate in high dimensions.

Error $= \sigma/\sqrt{N_s}$ for independent samples with finite variance. Variance and sampling cost can grow with d.

Our integral has three easy ingredients

$$\widehat{I} = \sum_{k=1}^{N_s} \underbrace{\frac{1}{N_s}}_{\text{equal weight}} \underbrace{f(x_k)}_{\text{estimator}}, \qquad f(x) = e^{-x^3}.$$

1. **Weight: easy.** Each sampled value enters with weight $1/N_s$. The sampling density is uniform: $p(x) = 1$ on $[0, 1]$.
2. **Sampling scheme: easy.** Generate independent uniform random numbers $x_k \in [0, 1]$.
3. **Estimator: easy.** Evaluate $f(x_k) = e^{-x_k^3}$ and average.

For a magnet, ask the same three questions: weights, sampling scheme, estimators.

N_s counts samples; N counts spins. The factor $1/N_s$ belongs to the sample average.

Classical Ising MCMC: the same three ingredients

For our square-lattice magnet, a configuration is a spin pattern z with energy $E(z) = -J \sum_{\langle i,j \rangle} z_i z_j$.

1. **Weight: Boltzmann.** $w(z) = e^{-\beta E(z)}$, with target probability $p(z) = w(z)/\mathcal{Z}$.
2. **Sampling scheme: Metropolis.** Pick a site uniformly, propose $z_i \mapsto -z_i$, then accept with

$$p_{\text{acc}} = \min\left(1, \frac{w(z')}{w(z)}\right) = \min(1, e^{-\beta \Delta E}), \quad \Delta E = E(z') - E(z).$$

3. **Estimators: read the pattern.** Record $E(z)$ or $m(z) = N^{-1} \sum_i z_i$. The classical Hamiltonian and these observables are diagonal in the spin basis.

Boltzmann weights set the visit frequency; average the recorded values with $1/N_s$.

Cluster updates can flip many spins together. Keep rejected moves; check equilibration and correlations.

Metropolis *et al.*, J. Chem. Phys. **21**, 1087 (1953).

Quantum spins: the two terms do not commute

$$H = \underbrace{-J \sum_i Z_i Z_{i+1}}_{H_z} \underbrace{-h \sum_i X_i}_{H_x}, \qquad [H_z, H_x] \neq 0.$$

Classical: one common spin basis

$$w(z) = e^{-\beta E(z)}$$

A pattern has a definite energy. The exponential acts on that number.

Quantum: spin flips mix patterns

$$w(z) = \langle z | \, e^{-\beta H} \, | z \rangle \neq e^{-\beta \langle z|H|z \rangle}$$

The weight depends on excursions through other spin states.

In general $e^{-\beta(H_z + H_x)} \neq e^{-\beta H_z} e^{-\beta H_x}$: we cannot just assign separate classical Boltzmann factors.

Noncommuting terms make the weight more than the energy of one spin pattern.

From a partition-function decomposition to QMC

$$\langle O \rangle = \frac{\mathrm{Tr}[O e^{-\beta H}]}{\mathrm{Tr}\, e^{-\beta H}} = \frac{\sum_{\mathcal{C}} W_{\mathcal{C}}\, O_{\mathcal{C}}}{\sum_{\mathcal{C}} W_{\mathcal{C}}}, \qquad \mathcal{Z} = \sum_{\mathcal{C}} W_{\mathcal{C}}.$$

0. **Configurations $\mathcal{C}$:** the objects defined by our chosen decomposition of $\mathcal{Z}$; not necessarily spin patterns.
1. **Weights $W_{\mathcal{C}}$:** each configuration's contribution to $\mathcal{Z}$, which we need to evaluate efficiently.
2. **Sampling scheme $\mathcal{C} \to \mathcal{C}'$:** moves that explore $p(\mathcal{C}) = W_{\mathcal{C}}/\mathcal{Z}$, when $W_{\mathcal{C}} \geq 0$.
3. **Estimators $O_{\mathcal{C}}$:** numbers computed from $\mathcal{C}$ whose weighted average gives $\langle O \rangle$.

Specify the configurations, weights, moves and estimators together.

What if the decomposition produces negative $W_{\mathcal{C}}$?

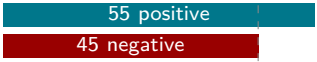
Negative weights require sign reweighting

If $W_{\mathcal{C}}$ can be negative, $W_{\mathcal{C}}/\mathcal{Z}$ is not a probability. Sample magnitudes and retain the sign:

$$p_{|W|}(\mathcal{C}) = \frac{|W_{\mathcal{C}}|}{\sum_{\mathcal{C}'} |W_{\mathcal{C}'}|}, \qquad s_{\mathcal{C}} = \operatorname{sgn}(W_{\mathcal{C}}).$$

$$\boxed{\langle O \rangle = \frac{\langle O_{\mathcal{C}} s_{\mathcal{C}} \rangle_{|W|}}{\langle s_{\mathcal{C}} \rangle_{|W|}}}, \qquad \bar{s} \equiv \langle s_{\mathcal{C}} \rangle_{|W|} = \frac{\sum_{\mathcal{C}} W_{\mathcal{C}}}{\sum_{\mathcal{C}} |W_{\mathcal{C}}|}.$$

A nearly fair coin: for $\bar{s} = 0.1$, 100 independent draws give, on average,



net sign $= 10$
but typical fluctuation $\approx \sqrt{100} = 10$!

The cancellation leaves a small signal buried in sampling noise.

Troyer & Wiese, Phys. Rev. Lett. **94**, 170201 (2005).

How costly is a small average sign?

The numerator is small too: $A \equiv \langle O_{\mathcal{C}} s_{\mathcal{C}} \rangle_{|W|} = \bar{s}\ \langle O \rangle$. **Dividing restores the value and amplifies the noise.**

Example: even if $\bar{s} = 0.1$ is known exactly

$$\widehat{\langle O \rangle} = \frac{0.100 \pm 0.010}{0.1} = 1.00 \pm 0.10.$$

To resolve $\langle O \rangle$ to ± 0.01:

need numerator error 0.001 instead of 0.010.

$10\times$ smaller error $\Rightarrow$ $100\times$ more samples.

With both numerator and sign sampled, the large-sample error is

$$\mathrm{SE}\left(\widehat{\langle O \rangle}\right) \simeq \frac{\sigma_{\mathrm{eff}}}{\bar{s}\sqrt{N_{\mathrm{eff}}}} \quad \Longrightarrow \quad \boxed{N_{\mathrm{eff}} \propto \bar{s}^{-2} \quad \text{at fixed error.}}$$

Hold σ_{eff} (observable noise and sign covariance) fixed; N_{eff} counts independent samples.

Sign penalty: $0.67 \to 2.2\times$, $0.1 \to 100\times$, $0.005 \to 40{,}000\times$ the samples.

A tenfold smaller average sign needs roughly 100 times more samples.

Troyer & Wiese, Phys. Rev. Lett. **94**, 170201 (2005). Example $\bar{s} = 0.67$: Ezzell, Barash & Hen, arXiv:2408.03924v3.

QMC examples

Suzuki: from Trotter slices to world lines

Let $H = A + B$, $A\,|z\rangle = E(z)\,|z\rangle$, and B be off-diagonal; $\Delta\tau = \beta/M$.

$$e^{-\beta H} = \left(e^{-\Delta\tau(A+B)}\right)^M \approx \Big(\underbrace{e^{-\Delta\tau A}}_{A_\Delta}\underbrace{e^{-\Delta\tau B}}_{B_\Delta}\Big)^M.$$

Trotter approximation at finite M; $\mathcal{Z} = \mathrm{Tr}\, e^{-\beta H} = \lim_{M\to\infty} \mathcal{Z}_M$.

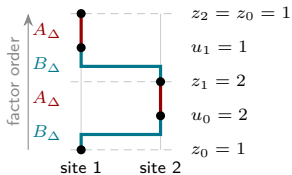
Insert $1 = \sum_z |z\rangle\langle z|$ into the trace:

$$\begin{aligned}
\mathcal{Z}_M &\equiv \mathrm{Tr}[(A_\Delta B_\Delta)^M] \\
&= \textstyle\sum_{z_0,\ldots,z_{M-1}} \prod_{\ell=0}^{M-1} w_\ell, \quad z_M = z_0. \\
w_\ell &\equiv \langle z_{\ell+1}|\, A_\Delta B_\Delta\, |z_\ell\rangle \\
&= \textstyle\sum_{u_\ell} \langle z_{\ell+1}|\, A_\Delta\, |u_\ell\rangle\langle u_\ell|\, B_\Delta\, |z_\ell\rangle \\
&= e^{-\Delta\tau E(z_{\ell+1})}\langle z_{\ell+1}|\, B_\Delta\, |z_\ell\rangle.
\end{aligned}$$

A: $u_\ell = z_{\ell+1}$, no change. B: kinks (or no hop).

M. Suzuki, Prog. Theor. Phys. 56, 1454–1469 (1976), doi:10.1143/PTP.56.1454.

One particle, one closed path



$M = 2$ example; read bottom to top.
Colors label factors, not particles.

SSE: weighted operator strings

The Taylor expansion turns the partition function into a sum over orders n:

$$Z = \operatorname{Tr} e^{-\beta H} = \sum_z \sum_{n=0}^{\infty} \frac{\beta^n}{n!} \langle z| (-H)^n |z\rangle .$$

Write $-H = \sum_{t,a} H_{t,a}$, where t is an operator type and a a site or bond. Each power is then a sum over **ordered strings of local operators**:

$$(-H)^n = \sum_{S_n} H_{t_n,a_n} \cdots H_{t_1,a_1}, \qquad S_n = ([t_1,a_1],\ldots,[t_n,a_n]).$$

The Taylor factor

$\beta^n/n!$ weights the expansion order.

The operator amplitude

$\langle z| H_{t_n,a_n} \cdots H_{t_1,a_1} |z\rangle$
weights the sequence of local actions.

An SSE configuration is a starting state and an operator string that returns to it.

Melko, *Stochastic Series Expansion QMC* (2013). Sec. 1.2.1, Eqs. 1.5–1.9.

SSE: a fixed-length operator string

Choose a maximum order M. Fill unused positions with $H_{0,0} = 1$.

For $n = 2$, $M = 3$, the same word (A, B) has three padded versions:

$$(A, B, 1), \qquad (A, 1, B), \qquad (1, A, B).$$

All $\binom{M}{n}$ copies have the same amplitude. Divide their Taylor weight:

$$\frac{\beta^n}{n!} \Big/ \binom{M}{n} = \frac{\beta^n (M-n)!}{M!}. \tag{1}$$

Summing over starting states z and all length-M strings S_M gives

$$\boxed{Z \approx Z_M = \sum_z \sum_{S_M} \frac{\beta^n (M-n)!}{M!} \langle z | H_{t_M, a_M} \cdots H_{t_1, a_1} | z \rangle .} \tag{2}$$

Here $n = n(S_M)$ counts Hamiltonian entries; $M - n$ entries are padding.

Sample (z, S_M) with this weight; increase M until the truncation is negligible.

Melko, *Stochastic Series Expansion QMC* (2013). Sec. 1.2.1, Eq. 1.10.

SSE example: the 2D TFIM

Square lattice, Pauli operators X_i, Z_i, and $J, h > 0$; work in the Z basis:

$$H = -J \sum_{\langle ij \rangle} Z_i Z_j - h \sum_i X_i = C1 - \sum_{t,a} H_{t,a}.$$

Operator	**Action**	**Weight**
$H_{1,\langle ij \rangle} = JZ_iZ_j + J1$	Keep aligned neighbours	$2J$ (else 0)
$H_{0,i} = h1$	Leave spin i unchanged	h
$H_{-1,i} = hX_i$	Flip spin i	h

Why $+J$? Shifting $\pm J \to \{0, 2J\}$ removes negative bond weights and avoids the sign problem.

Two flips: $(hX_i, 1, hX_i)$ returns to z; $n = 2$, $M = 3$, weight $\beta^2 h^2/6$.

Energy shift: $C = JN_b + hN$, $Z \approx e^{-\beta C} Z_M$ (N sites, N_b bonds).

Heisenberg antiferromagnets: local spin rotations also needed (appendix details).

Melko, *Stochastic Series Expansion QMC* (2013). Sec. 1.4, Eqs. 1.31–1.38.

Stochastic series expansion (SSE)

Annoyance: Choice of H decomposition is ad-hoc, system specific, and changes details of QMC sampling

Spin-1/2 Heisenberg SSE-QMC : A.W. Sandvik, Phys. Rev. B 56, 11678 (1997)

Spin-1/2 TFIM SSE-QMC: A.W. Sandvik, Phys. Rev. E 68, 056701 (2003)

Spin-1/2 Rydberg array SSE-QMC: E. Merali, I.J.S. De Vlugt, R.G. Melko, SciPost Phys. Core 7, 016 (2024)

⋮

PMR-QMC: expand about the classical limit

Split $H = D_0 + V$: diagonal energies in D_0, off-diagonal moves in V.

$$\boxed{Z = Z_{\mathrm{cl}} + \textstyle\sum_{q>0} Z_q}, \qquad Z_{\mathrm{cl}} = Z_0 = \textstyle\sum_z e^{-\beta E_z}.$$

Z_q collects closed sequences of q moves; $E_z = \langle z| D_0 |z\rangle$.

q counts off-diagonal moves: the auxiliary "quantum dimension".

One-spin example

$$H = -\sigma^z - h\sigma^x, \quad D_0 = -\sigma^z.$$

$$Z_{\mathrm{cl}} = 2\cosh\beta,$$

$$Z = 2\cosh\left(\beta\sqrt{1+h^2}\right).$$

At $h = 0$: $Z = Z_{\mathrm{cl}}$ and sampling reduces to classical Monte Carlo.

Exact values for $h = 1$

β	Z_{cl}	$\sum_{q>0} Z_q$	Z
0.5	2.255	0.266	2.521
1.0	3.086	1.270	4.356
2.0	7.524	9.454	16.978

Energy unit: coefficient of σ^z.

Gupta, Albash & Hen, J. Stat. Mech. (2020) 073105. Numbers from the exact one-spin spectrum.

PMR-QMC covers arbitrary spin systems

The generalization to higher spins is already available.

A quantum Monte Carlo algorithm for arbitrary high-spin Hamiltonians

A. Babakhani, L. Barash & I. Hen

Computer Physics Communications **321**, 110037 (2026)

Spin s has $2s+1$ local states. The method handles arbitrary connectivity, including long-range and multi-spin interactions.

For today's examples: use the earlier spin-$\frac{1}{2}$ version.

Two local states make the moves easy to follow:

$$|0\rangle \xleftrightarrow{X} |1\rangle\,, \qquad X^2 = 1.$$

We can see amplitudes, closed paths, and quantum corrections with simple spin flips.

Spin-$\frac{1}{2}$ keeps the explanation simple; the PMR-QMC framework extends beyond it.

Spin-$\frac{1}{2}$ method: Barash, Babakhani & Hen, Phys. Rev. Research **6**, 013281 (2024).

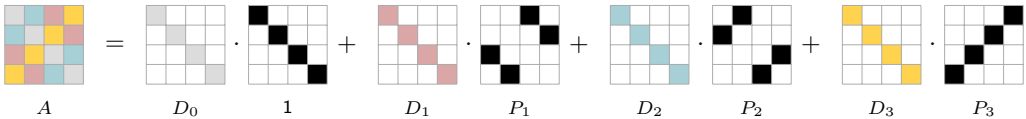
Permutation Matrix Representation (PMR)

PMR separates amplitudes from moves

For any finite $d \times d$ matrix—Hamiltonian or observable—we can write

$$\boxed{A = \sum_{g \in G} D_g \, P_g}, \qquad D_g P_g \, |y\rangle = \underbrace{(D_g)_{g(y),g(y)}}_{\text{amplitude at the destination}} \; |g(y)\rangle \, .$$

D_g: a diagonal matrix. **P_g: a permutation of basis states.**



The group G organizes the moves so that each matrix entry belongs to exactly one term.

A suitable group always exists. It gives unique amplitudes and a simple loop test.

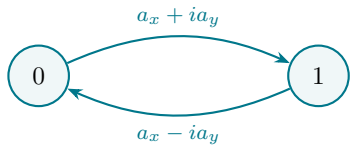
Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Matrix graphic: PMR-QMC methods deck, PDF page 9.

One spin: the only move is a flip

The computational basis $\{|0\rangle, |1\rangle\}$ diagonalizes Z: $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$.

$$A = a_0 1 + a_x X + a_y Y + a_z Z, \qquad Y = -iZX.$$

$$A = \underbrace{(a_0 1 + a_z Z)}_{D_e} 1 + \underbrace{(a_x 1 - i a_y Z)}_{D_x} X, \qquad G = \{e, x\} \cong C_2.$$



$$A = \begin{pmatrix} a_0 + a_z & a_x - i a_y \\ a_x + i a_y & a_0 - a_z \end{pmatrix}.$$

The phase from Y lives in D_x;
X only changes the state.

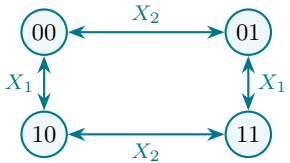
Four scalar Pauli coefficients become two diagonal-matrix coefficients.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Sec. III.2.1.

Two spins reveal a choice of PMR group

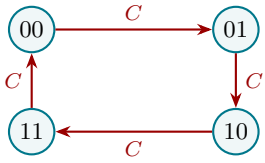
Independent bit flips: $C_2 \times C_2$.

$$G_X = \{1, X_1, X_2, X_1X_2\}.$$



One global cycle: C_4.

$$G_C = \{1, C, C^2, C^3\}.$$



For n spins, $G_X^{(n)} \cong (C_2)^n$: the bit mask s specifies the unique move $P_s = \prod_{j=1}^{n} X_j^{s_j}$ sending $|z\rangle \to |z \oplus s\rangle$.

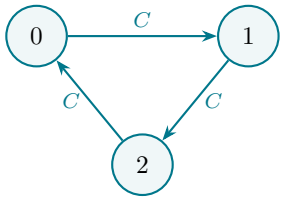
The group can vary. The coefficients are unique only after that choice.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Secs. III.1–III.2.

A cyclic shift proves existence in every dimension

On basis labels $0, \ldots, d-1$, define

$$C\,|y\rangle = |y+1 \bmod d\rangle\,, \qquad G = \{1, C, \ldots, C^{d-1}\} \cong C_d.$$



$d = 3$ example

The unique shift $y \to x$ is $k = x - y \pmod d$.
Give it the amplitude

$$(D_k)_{xx} = A_{x,\, x-k \bmod d}.$$

Then $A = \sum_{k=0}^{d-1} D_k C^k$ is an Abelian PMR.

Existence is constructive: each matrix entry has one shift and one amplitude.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Existence theorem; see also Gupta, Albash & Hen, J. Stat. Mech. (2020) 073105, App. A.

Regular action means one move per ordered pair

Choose a permutation group G acting on the d basis labels.

Transitive: every state is reachable.

$$\forall\, x, y, \quad \exists\, g \in G :\ g(y) = x.$$

Free: only the identity fixes a state.

$$g(y) = y \text{ for some } y \quad \Longrightarrow \quad g = e.$$

Together: regular (simply transitive).

$$\boxed{\forall\, x, y, \quad \exists!\, g \in G :\ g(y) = x.}$$

If $g(y) = h(y)$, then $h^{-1}g$ fixes y; freeness forces $g = h$.

For fixed y, group elements label all destinations exactly once: $|G| = d$.

Regularity is the PMR requirement. Commutativity is optional.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Definition of PMR form.

Read off the diagonals by undoing the move

Define $\mathrm{Diag}(B)$ to keep the diagonal of B and set its other entries to zero.

$$\boxed{D_g = \mathrm{Diag}(AP_g^{-1}) = \mathrm{Diag}(AP_g^{T})}, \qquad (D_g)_{xx} = A_{x,\, g^{-1}(x)}.$$

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

$$D_0 = \mathrm{diag}(a, e, i),$$
$$D_1 = \mathrm{diag}(c, d, h),$$
$$D_2 = \mathrm{diag}(b, f, g),$$
$$A = D_0 + D_1 C + D_2 C^2.$$

Once the basis and group are fixed, every D_g is uniquely determined.

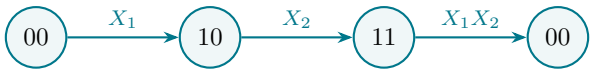
Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Theorem on computing the diagonal coefficients.

A return to one state forces the identity

Let $S = P_{g_q} \cdots P_{g_1}$ be any ordered string from a regular PMR group.

$$\boxed{S\,|z\rangle = |z\rangle \text{ for one basis state } z \quad \Longleftrightarrow \quad S = 1.}$$

Proof in one line: closure puts S in the group; freeness rules out a fixed point for $S \neq 1$.



$$S = (X_1X_2)X_2X_1 = 1$$

For Pauli-X strings, the test is just $s_1 \oplus \cdots \oplus s_q = 0$.

Permutation closure is independent of the starting state. The weight is not.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Fixed-point corollary.

PMR-QMC partition function

Build quantum corrections on a classical starting point

Our target for $H = D_0 + \sum_{j \neq e} D_j P_j$ is

$$\boxed{Z = Z_{\rm cl} + \sum_{q>0} Z_q}, \qquad Z_{\rm cl} = \sum_z e^{-\beta E_z}.$$

Keep classical physics exact

Even with no moves, SSE has infinitely many diagonal terms:

$$\sum_{n=0}^{\infty} \frac{(-\beta E_z)^n}{n!} = e^{-\beta E_z}.$$

Resum them. For $q > 0$, also sum every diagonal insertion along the path.

Sample only the quantum moves

The trace requires a closed path. The PMR group gives its test:

$$P_{j_q} \cdots P_{j_1} = 1.$$

Sample z and the ordered moves; q is the auxiliary "quantum dimension".

Recall $H = -\sigma^z - \sigma^x$**:** $\beta = 1$ gives $Z = 3.086 + 1.270$ (classical + quantum).

Remove the off-diagonal terms: only $q = 0$ remains, giving classical Monte Carlo.

Gupta, Albash & Hen, J. Stat. Mech. (2020) 073105. Ezzell & Hen, arXiv:2504.07295v2. Sec. V.1.

SSE recap: what PMR-QMC changes

For $-H = \sum_{t,a} H_{t,a}$, SSE sums weighted operator strings:

$$Z = \sum_{z} \sum_{n=0}^{\infty} \sum_{S_n} \frac{\beta^n}{n!} \langle z | H_{t_n,a_n} \cdots H_{t_1,a_1} | z \rangle .$$

Padding to M slots gives $\frac{\beta^n}{n!} \Big/ \binom{M}{n} = \frac{\beta^n (M-n)!}{M!}$ for $n \le M$.
PMR writes $H = D_0 + \sum_{j \neq e} D_j P_j$, with $D_0 \equiv D_e$ diagonal.

(a) Group structure

Factor each hop as $D_j P_j$. Regularity turns return to z into the identity test:

$$S|z\rangle = |z\rangle \iff S = 1, \qquad S = P_{j_q} \cdots P_{j_1}.$$

(b) Exact resummation

Keep the same ordered q moves. Sum over **all numbers and positions** of D_0 insertions, with their Taylor weights.

PMR samples $(z; j_1, \ldots, j_q)$; diagonal insertions are already summed into its weight.

Melko, *Stochastic Series Expansion QMC* (2013). Sec. 1.2.1. Ezzell & Hen, arXiv:2504.07295v2. Secs. III and V.1.

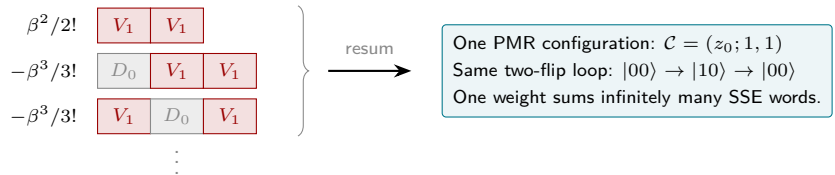
The off-diagonal expansion is a resummation

Let $H = D_0 + \sum_j V_j$, $V_j = D_j P_j$. Group the SSE words C_n by their hops:

$$\langle z|\, e^{-\beta H}\, |z\rangle = \sum_{n=0}^{\infty} \sum_{C_n} \frac{(-\beta)^n}{n!} \langle z|\, C_n\, |z\rangle$$

$$= \sum_{q=0}^{\infty} \sum_{\mathbf{i}_q} D_{\mathbf{i}_q}(z)\, \langle z|\, S\, |z\rangle\ R_\beta(E_0, \ldots, E_q).$$

$$R_\beta(E_0, \ldots, E_q) = \sum_{n=q}^{\infty} \frac{(-\beta)^n}{n!} \quad \sum_{k_0+\cdots+k_q=n-q}\ \prod_{\ell=0}^{q} E_\ell^{k_\ell}.$$



$D_{\mathbf{i}_q}(z) = \prod_{\ell=1}^{q} (D_{i_\ell})_{z_\ell z_\ell}$, $S = P_{i_q} \cdots P_{i_1}$; $E_\ell = (D_0)_{z_\ell z_\ell}$, $k_\ell \geq 0$.

Ezzell & Hen, arXiv:2504.07295v2. Sec. V.1.

The missing ingredient: divided differences

To evaluate R_β, use the **divided difference**: a generalization of a secant slope.

$$f[x_0] = f(x_0), \qquad f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

For distinct nodes, higher orders are defined recursively:

$$f[x_0, \ldots, x_q] = \frac{f[x_1, \ldots, x_q] - f[x_0, \ldots, x_{q-1}]}{x_q - x_0}.$$

Repeated nodes use continuous limits. If all $q+1$ nodes coincide,

$$f[x, \ldots, x] = \frac{f^{(q)}(x)}{q!}.$$

The resummation is exactly this object for the exponential:

$$\boxed{R_\beta(E_0, \ldots, E_q) = f_\beta[E_0, \ldots, E_q]}, \qquad f_\beta(E) = e^{-\beta E}.$$

Ezzell & Hen, arXiv:2504.07295v2. Sec. IV (review of divided differences), Sec. V.1 (off-diagonal expansion).

The divided difference

For $f(x) = e^{tx}$, the shorthand $e^{t[x_0,\ldots,x_q]}$ means $f[x_0,\ldots,x_q]$.

$$f[x_0,\ldots,x_q] = \frac{1}{2\pi i}\oint_\Gamma \frac{f(x)}{\prod_{i=0}^{q}(x-x_i)}\,\mathrm{d}x = \sum_{m=0}^{\infty}\frac{f^{(q+m)}(0)}{(q+m)!}\sum_{k_0+\cdots+k_q=m}\prod_{j=0}^{q}x_j^{k_j}.$$

x_i distinct

$x_i = x$ for every i

Polynomial interpolant form

$$f[x_0,\ldots,x_q] = \sum_{i=0}^{q}\frac{f(x_i)}{\prod_{k\neq i}(x_i-x_k)}.$$

Taylor coefficient

$$f[x,\ldots,x] = \frac{f^{(q)}(x)}{q!}.$$

Recursive form

$$f[x_0] = f(x_0), \qquad f[x_0,x_1] = \frac{f(x_1)-f(x_0)}{x_1-x_0}, \quad \ldots$$

IUPAP CCP 2025, "The divided difference" (printed slide 59). Ezzell & Hen, arXiv:2504.07295v2. Sec. IV.

Stable, incremental evaluation

2020: Gupta, Barash & Hen developed an accurate method to update exponential divided differences as energy inputs are added or removed.

- **Shift and scale.** Bring the inputs into a range where the exponential can be evaluated accurately, including nearby or repeated energies.
- **Reuse intermediate results.** Keep the existing calculation in memory and update it when one energy is added or removed.

$$\text{Work per input update:} \qquad O(sq) = O(q) \quad \text{at fixed } s.$$

Here $q + 1$ is the number of energies; the scaling s depends on their spread in $-\beta E$.

Analytical Results

What can contribute to a thermal average?

$$H = \sum_{P \in S_H} D_P P, \qquad A = \sum_{P \in G} \Lambda_P P, \qquad K = \langle S_H \rangle \leq G.$$

In the common group G, $S_H = \{P : D_P \neq 0\}$ lists active moves; K includes all products.

Observable move P	**Consequence for** $\langle \Lambda_P P \rangle$
$P \in S_H$	Already in H: direct formal estimator.
$P \in K \setminus S_H$	Generated: use products of Hamiltonian terms.
$P \notin K$	Exactly zero: no Hamiltonian path can close.

For the first two cases, use a canonical representation to handle possible zeros in the Hamiltonian amplitudes.

Group membership gives a selection rule; it does not guarantee a nonzero average.

Ezzell & Hen, arXiv:2504.07295v2, Secs. VI.5–VI.7.3.

The observable must close a Hamiltonian path

Every Hamiltonian string has permutation $S \in K$. A contribution to the trace requires

$$\langle z|\, PS\, |z\rangle \neq 0 \quad \Longrightarrow \quad PS = 1 \quad \Longrightarrow \quad P = S^{-1} \in K.$$

Three spins with pair flips

$$H = -J(X_1X_2 + X_2X_3), \quad J > 0,$$
$$K = \{1, X_1X_2, X_2X_3, X_1X_3\}.$$

X_1X_2: already in H.

$X_1X_3 = (X_1X_2)(X_2X_3)$: generated.

$X_1 \notin K$: $\langle \Lambda X_1 \rangle = 0$
for every diagonal Λ.

All moves in K flip an even number of bits.

$$\boxed{\langle A \rangle = \langle A_K \rangle, \qquad A_K = \textstyle\sum_{P \in K} \Lambda_P P.}$$

Discard the forbidden components analytically; estimate the remaining ones.

Ezzell & Hen, arXiv:2504.07295v2, Sec. VI.7.3; pair-flip illustration.

Arbitrary observables have PMR-QMC estimators

For any finite-dimensional operator A, the construction gives an unbiased estimator of

$$\boxed{\langle A \rangle = \frac{\mathrm{Tr}(Ae^{-\beta H})}{Z} = \frac{\sum_{\mathcal{C}} W_{\mathcal{C}} A_{\mathcal{C}}}{\sum_{\mathcal{C}} W_{\mathcal{C}}}}.$$

1. **Use the group selection rule.** Discard components that cannot close a Hamiltonian path.
2. **Build a canonical form.** Combine diagonal factors with products of actual Hamiltonian terms, respecting their zero amplitudes.
3. **Measure and combine.** Evaluate each term on PMR configurations and combine its contributions.

The framework also supplies estimators for moments, imaginary-time correlations, and their integrals.

Change the observable to change the physics question, using the same sampled paths.

Ezzell & Hen, arXiv:2504.07295v2. Secs. VI.5–VI.8 and VIII. $W_{\mathcal{C}}$ is the PMR weight; $A_{\mathcal{C}}$ is the observable estimator.

A Hamiltonian term has a simple estimator

For $H = D_0 + \sum_{l \neq 0} D_l P_l$, a sampled path is $\mathcal{C} = (z_0; i_1, \ldots, i_q)$, with $E_j = \langle z_j | \, D_0 \, | z_j \rangle$.

Diagonal energy: read the starting-state energy,

$$(D_0)_{\mathcal{C}} = E_0.$$

Off-diagonal term: select paths whose last move is P_l,

$$(D_l P_l)_{\mathcal{C}} = \begin{cases} \dfrac{f_\beta[E_0, \ldots, E_{q-1}]}{f_\beta[E_0, \ldots, E_q]}, & q \geq 1, \; i_q = l, \\ 0, & \text{otherwise}, \end{cases} \qquad f_\beta(E) = e^{-\beta E}.$$

The ratio removes the last energy. Its Monte Carlo average gives $\langle D_l P_l \rangle$.

Linearity then gives the expectation of any sum of Hamiltonian terms.

IUPAP CCP 2025, PDF page 69, "Terms that comprise H". Ezzell & Hen, arXiv:2504.07295v2. Secs. VI.1 and VI.4.

Dynamic estimators summary

We extend the static estimators for A and B to

1. **Imaginary-time correlations:** $\langle A(\tau)B\rangle = \left\langle e^{\tau H} A e^{-\tau H} B\right\rangle.$
2. **Integrated correlations:** $\displaystyle\int_0^\beta \langle A(\tau)B\rangle \, \mathrm{d}\tau.$
3. **Time-weighted integrals:** $\displaystyle\int_0^{\beta/2} \tau \, \langle A(\tau)B\rangle \, \mathrm{d}\tau.$

For (1): a noncommutative generalization of the Leibniz rule.

From (1) to (2) and (3): divided-difference identities evaluate these integrals analytically.

The same framework measures static averages and their imaginary-time response.

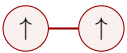
IUPAP CCP 2025, "Dynamic estimators summary" (printed slide 69, PDF page 71). Ezzell & Hen, arXiv:2504.07295v2. Sec. VIII.

Numerical Results

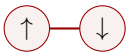
Two spins: a field competes with their bond

$$H(\lambda) = \underbrace{Z_1 Z_2}_{\text{prefer opposite spins}} + \underbrace{0.1(X_1 + X_2)}_{\text{mix spin configurations}} + \underbrace{\lambda(Z_1 + Z_2)}_{\text{favor one direction}} .$$

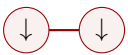
Large negative λ



Small $|\lambda|$



Large positive λ



Without the X term:

Opposite spins: energy -1. Aligned spins: $1 \pm 2\lambda$.

These cross at $\lambda = \pm 1$.

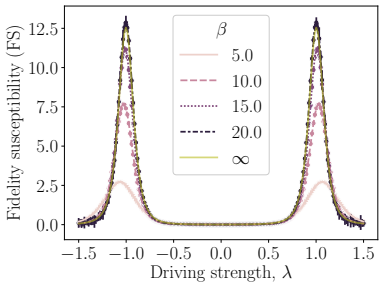
With spin mixing:

The ground state changes smoothly through an avoided crossing.

A small, exactly solvable test of detecting rapid ground-state change.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Eq. (1). Arrows show the preferred Z-basis patterns before mixing.

The response peaks find both avoided crossings



Fidelity susceptibility (FS)

How rapidly does the ground state change when we turn λ?

Result

QMC matches exact finite-temperature curves. Peaks sit near $\lambda = \pm 1$.

At $\beta = 20$, FS is close to the ground-state result.

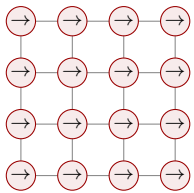
$\beta = 1/T$: larger β means colder.

We can probe $H_1 = Z_1 + Z_2$, even though it is only part of the diagonal energy.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 1. Points: QMC over 5 independent runs; lines: exact calculations.

Ising magnet: alignment versus quantum fluctuations

$$H = H_J + hH_h, \qquad H_J = -\sum_{\langle i,j \rangle} X_i X_j, \qquad H_h = -\sum_i Z_i.$$



4×4, periodic boundaries
(wraparound bonds omitted)

Bonds align spins along x; the field favors z.

Cool at $h = 1$. Compare energy susceptibility (ES) and FS for bond and field perturbations.

Keep positive parity: $\prod_i Z_i = +1$.

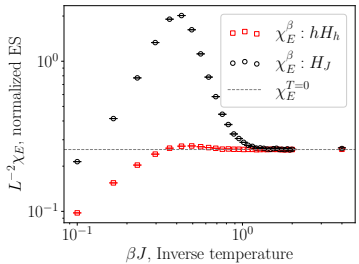
Only an even number of down spins in each sampled Z-basis state.

This benchmark checks cooling and symmetry-sector control at a fixed field.

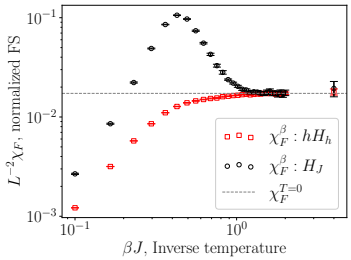
Ezzell, Barash & Hen, arXiv:2408.03924v3. Eq. (2), Fig. 2 and Appendix C. Diagram: alignment tendency in the x direction.

Different thermal responses meet as the magnet cools

Energy susceptibility



Fidelity susceptibility



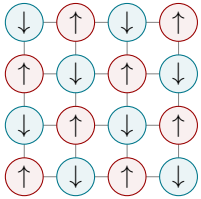
Black: perturb bonds. Red: perturb the field. 4×4, $h = 1$, positive parity.

At finite T the choice matters; at low T both approach the same response here.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 2. 100 independent runs. FS is consistent with Albuquerque et al., PRB 81, 064418 (2010).

XXZ magnet: a checkerboard that can exchange spins

$$H = \frac{J_z}{4} \sum_{\langle i,j \rangle} Z_i Z_j + \frac{\lambda}{4} \sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j).$$



Easy-axis Néel order (schematic)

ZZ: prefer opposite neighbors.

$XX + YY$: exchange $\uparrow\downarrow \leftrightarrow \downarrow\uparrow$.

For $J_z > \lambda$, checkerboard order wins at low temperature.

Our scan: $J_z = 1.5$, $\lambda = 1$; heat the easy-axis magnet.

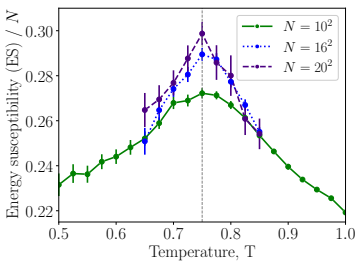
Known thermal transition: $T_c \simeq 0.75$.

The numerical question: can ES and FS see the checkerboard order melt?

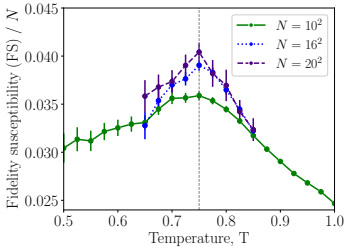
Ezzell, Barash & Hen, arXiv:2408.03924v3. Eq. (4), Fig. 3. Pauli convention; $k_B = 1$ and energies in units of $\lambda = 1$.

Larger XXZ lattices sharpen the thermal peak

Energy susceptibility per spin



Fidelity susceptibility per spin

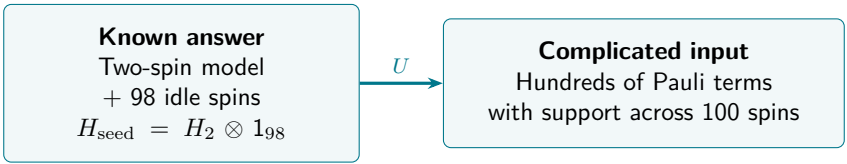


$N = 100,\ 256,\ 400$ spins. Both responses develop peaks near $T \simeq 0.75$.

The same estimators reach lattice sizes that reveal a collective thermal transition.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 3. At least 200 independent runs; FS agrees with Wang et al., PRX 5, 031007 (2015).

Hide a simple model inside 100 spins



$$H_U = U H_{\mathrm{seed}} U^\dagger, \qquad (H_1)_U = U(H_1 \otimes 1_{98}) U^\dagger.$$

Rotate the model and its probe together: the spectrum and corresponding thermal averages stay the same.

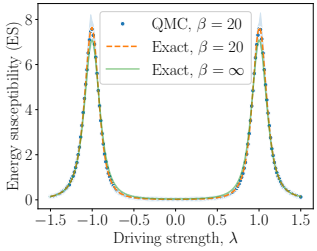
The universal study generated 50 candidates and kept the 10 with the best average signs in a pilot run.

A controlled test of complicated inputs, with an exact answer still available.

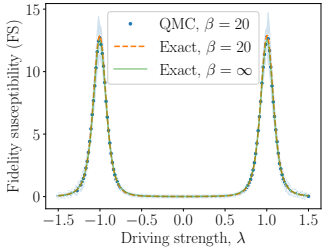
Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4 and Appendix D. Screening at $\beta = 5$, $\lambda = 1$.

The rotated ensemble reproduces the response peaks

Energy susceptibility



Fidelity susceptibility

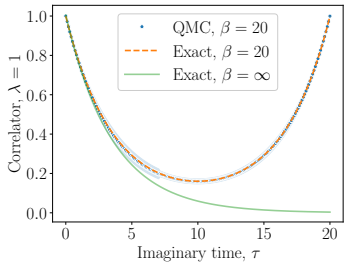


100-spin ensemble, $\beta = 20$: QMC agrees with the exact finite-temperature curves.

FS is sharper here; ES already locates both rapid-change regions.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4(b,c). Averages over 10 selected models. Dashed: exact finite β; green: ground state.

The full imaginary-time response also agrees



Rotated ensemble; $\lambda = 1$, $\beta = 20$.

Follow fluctuations in imaginary time.

$$G(\tau) = \langle V(\tau)V\rangle - \langle V\rangle^2, \qquad V = (H_1)_U.$$

ES integrates G; FS weights longer times:

$$\chi_E = \int_0^{\beta} G(\tau)\,\mathrm{d}\tau,$$

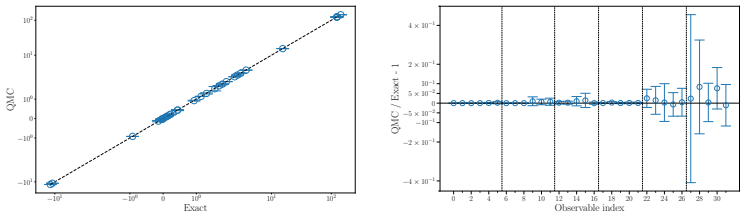
$$\chi_F = \int_0^{\beta/2} \tau G(\tau)\,\mathrm{d}\tau.$$

QMC captures the response curve behind the two susceptibility numbers.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 4(a), Eqs. (19)–(21). Imaginary time is an equilibrium probe, not real-time motion.

First, check 32 measurements against exact answers

3×3 **Ising magnet:** $H = -\sum_{\langle i,j \rangle} Z_i Z_j - \lambda \sum_i X_i, \quad \beta = 1,\ \lambda = 0.5.$



Tested: energies, moments, correlations, susceptibilities, and custom probes such as $A = X_1 + Z_2 Z_3$.

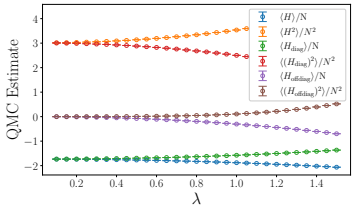
All 32 plotted quantities agree with exact numerical linear algebra within 2σ.

Ezzell & Hen, arXiv:2504.07295v2. Fig. 1 (indices 0–31). Open boundaries; 100 independent runs; displayed bars are 3σ.

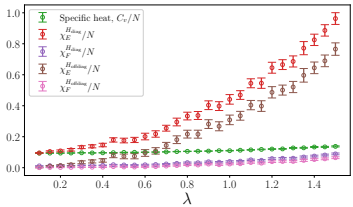
On 64 spins, one run measures a broad response panel

8×8 **TFIM, open boundaries:** $\beta = 1$, sweep $\lambda = 0.1$ to 1.5.

Energy and its moments



Heat capacity, ES and FS



As $\lambda \to 0$**:** the off-diagonal energy vanishes and $\langle H \rangle \to \langle H_{\text{diag}} \rangle$.

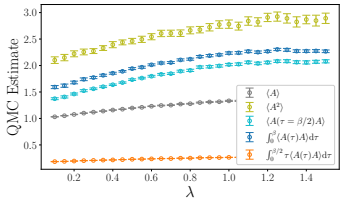
The same simulation supplies static averages and fluctuation-based responses.

Ezzell & Hen, arXiv:2504.07295v2. Fig. 2(a,c). 100 independent runs; 2σ bars. Correlation measurements also collected in the same runs.

Custom probes can combine local and distant spins

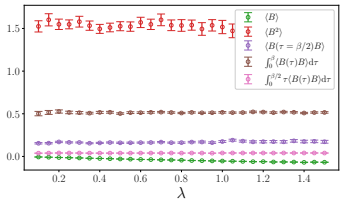
One spin flip + a spin-pair probe

$$A = X_1 + Z_2 Z_3$$



A sum of three selected Pauli strings

$$B \simeq 0.93 Y_{62} + 0.78 Y_{17} Z_{53} - 0.24 Z_{22} X_{31} Z_{49}$$



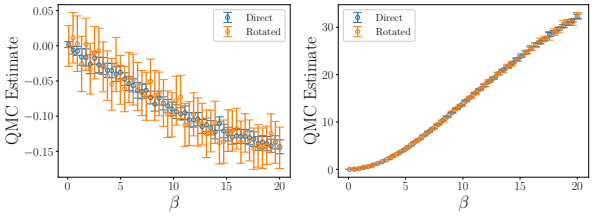
Means, second moments and time correlations work for probes beyond the energy.

Ezzell & Hen, arXiv:2504.07295v2. Fig. 3(a,b), Eqs. (140)–(141). Same 8×8 TFIM, $\beta = 1$; 100 runs, 2σ bars.

Rotate a custom observable: the answer is preserved

Same solvable seed, $\lambda = 1$**:** $H_U = UHU^\dagger, \quad O_U = UOU^\dagger.$

Mean $\langle O \rangle$ **Weighted integral** $\int_0^{\beta/2} \tau \langle O(\tau) O \rangle \, \mathrm{d}\tau$



H_U: **179 Pauli terms**. O_U: **295 terms**, expanded from 5.

Both measurements agree before and after spreading the model across 100 spins.

Ezzell & Hen, arXiv:2504.07295v2. Fig. 4, Eqs. (142)–(144). One selected sign-problem-free rotated instance; 3σ bars.

Conclusion

Questions?

Appendix / backup

Backup: Heisenberg SSE decomposition

Spin-1/2 antiferromagnet, $H = \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$, in the S^z basis:

$$H_{0,0} = 1, \qquad H_{1,a} = \tfrac{1}{4} - S_i^z S_j^z, \qquad H_{2,a} = \tfrac{1}{2}\left(S_i^+ S_j^- + S_i^- S_j^+\right).$$

1. An offset, added by hand.
The $\frac{1}{4}$ is there to stop $H_{1,a}$ from having a negative matrix element. It shifts the energy by $N_b/4$, which the estimator has to put back.

2. A rotation of the basis.
One sublattice is turned by π about S^z, so $S^\pm \to -S^\pm$ and $H_{2,a}$ comes out positive. Bipartite lattices only: the general version *is* the sign problem.

Then every surviving matrix element is $\frac{1}{2}$ — and the TFIM needs a different set:
$H_{-1,a} = h(\sigma_a^+ + \sigma_a^-)$, $H_{0,a} = h$, $H_{1,a} = J(\sigma_i^z \sigma_j^z + 1)$, with a a site in one and a bond in the other.

These decomposition choices must be worked out for each model.

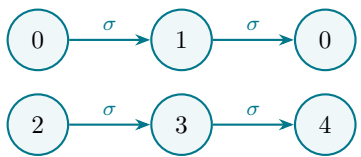
Melko, *Stochastic Series Expansion QMC* (2013). Heisenberg: Eqs. 1.18–1.20 and the two remarks after them. TFIM: Eqs. 1.32–1.35.

Group theory of PMR

Fixed-point-free moves are not enough

Take $\sigma = (01)(234)$ on five basis states. Both σ and σ^{-1} move every state.

Apply σ twice.



$$\sigma^2 = (243) \neq e.$$

It fixes 0 and 1, but moves $2, 3, 4$.

$$\langle 0 | P_\sigma^2 | 0 \rangle = 1, \qquad P_\sigma^2 \neq 1.$$

So a return to one state need not mean an identity permutation.

The condition must hold for the generated group, including all products.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Counterexample following the fixed-point corollary.

Backup: coefficient extraction is a projection

$\mathrm{Diag}(B)$ keeps the diagonal of B and sets its other entries to zero.

For a regular PMR group, the permutation supports are disjoint:

$$\mathrm{Diag}(P_h P_g^{-1}) = \delta_{hg} \mathbf{1}.$$

Since left multiplication by a diagonal matrix only rescales rows,

$$\begin{aligned}\mathrm{Diag}(A P_g^{-1}) &= \sum_{h \in G} D_h \,\mathrm{Diag}(P_h P_g^{-1}) \\ &= D_g.\end{aligned}$$

Taking the trace gives scalar orthogonality, $\mathrm{Tr}(P_g^\dagger P_h) = d\,\delta_{gh}$, but

$$\mathrm{Tr}(A P_g^{-1}) = \mathrm{Tr}(D_g) \quad \text{retains only the sum of its diagonal entries.}$$

Diagonal projection recovers all d amplitudes; a scalar trace cannot.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Proof of the coefficient-extraction theorem.

Backup: regular groups can be non-Abelian

Take any finite group K of order d and label the basis states by $h \in K$.

$$L_g \left|h\right\rangle = \left|gh\right\rangle, \qquad L_g L_k = L_{gk}.$$

Transitive.

To send $|h\rangle$ to $|k\rangle$, choose

$$g = kh^{-1}.$$

Free.

A fixed basis state implies

$$gh = h \quad \Longrightarrow \quad g = e.$$

Example: $K = S_3$ gives six permutation matrices acting regularly on **six** basis states.

Its ordinary action on three labels is a different action and is not free.

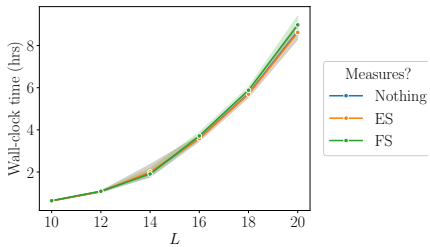
Regularity concerns the action. Abelianity concerns multiplication within the group.

Ezzell & Hen, arXiv:2504.07295v2, Sec. III. Sec. III.1; explicit left-regular-action construction.

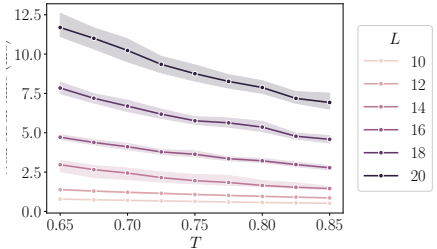
Simulation timing

XXZ timings: sparse measurements add little overhead

Increase the lattice size



Cool at fixed lattice size



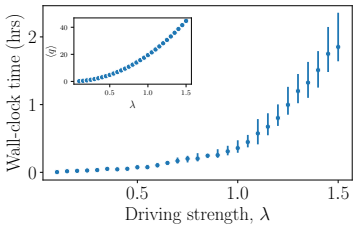
At $T \simeq T_c$, measured time grows roughly as $L^4 = N^2$ in this benchmark.

Measuring every 5,000 moves makes sampling dominate the total runtime here.

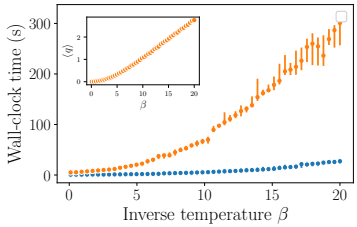
Ezzell, Barash & Hen, arXiv:2408.03924v3. Fig. 6. Fixed 4×10^8 moves, no thermalization in this timing test; 20 runs on AMD EPYC 7513.

Measurement breadth has a visible computational cost

64-spin TFIM: all observables



Two-spin versus rotated model



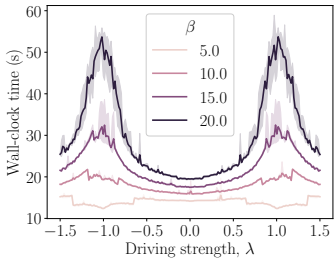
More field-induced flips increase effort. Rotating O from **5 to 295 terms** adds measurement work even at similar mean string length $\langle q \rangle$.

Runtime depends on both the sampled paths and the observable we ask to measure.

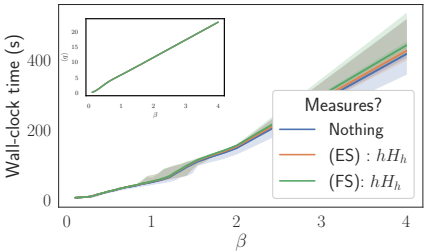
Ezzell & Hen, arXiv:2504.07295v2. Fig. 5. Left: 100 parallel runs; right: 6. Timing bars show empirical 95% intervals.

Backup: sampling costs vary across parameters

Two spins: cost peaks at the crossings



4×4 TFIM: time grows on cooling



Two-spin production runs: roughly **10 s–1 min**. TFIM timing fit: $T_{\text{wall}} \propto \beta^{1.23}$.

Simple spectra do not imply uniform sampling cost across parameters.

Ezzell, Barash & Hen, arXiv:2408.03924v3. Figs. 7 and 10, Table I. Empirical timings with the paper's fixed simulation protocols.